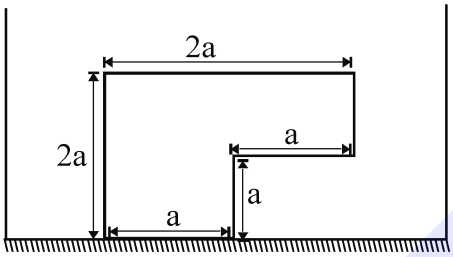
SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. A piece of material whose specific gravity is 'S' is placed in a container as shown in figure and water is poured slowly, then piece will



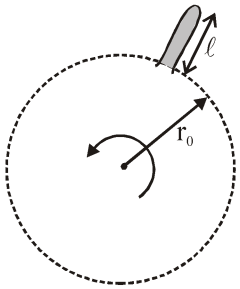
(A) topple clockwise if $S < \frac{1}{2}$.

(B) topple anticlockwise if $S < \frac{1}{2}$.

(C) topple clockwise if S > 2.

(D) topple anticlockwise if S > 2.

2. A test tube filled with water is being spun around in an ultracentrifuge with angular velocity ω. The test tube is lying along a radius and the free surface of the water is at radius r_0 (Figure). The length of the test tube is ℓ and ρ is the density of the water. Ignore gravity and ignore atmospheric pressure. Choose the correct statement.



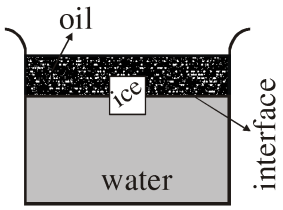
(A) The pressure at the mid point of test tube is given by $\frac{1}{8}\rho\ell\omega^2(\ell + 4r_0)$.

(B) The pressure at the mid point of test tube is given by $\frac{1}{2}\rho\ell\omega^2\left(\frac{\ell}{2} + r_0\right)$.

(C) The pressure at the mid point of test tube is given by $\frac{1}{4}\rho\ell\omega^2\left(\frac{\ell}{2} + r_0\right)$.

(D) The pressure remains constant through out the tube.

3. An ice cube is floating in water above which a layer of a lighter oil is poured. As the ice melts completely, the level of interface and the upper most level of oil will respectively :-



(A) rise and fall

(B) fall and rise

(C) rise and not change

(D) not change and fall

4. A thin uniform cylindrical shell, closed at both ends, is partially filled with water. It is floating vertically in water in half-submerged state. If ρ_C is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is **[IIT-JEE-2012]**

(A) more than half-filled if ρ_C is less than 0.5

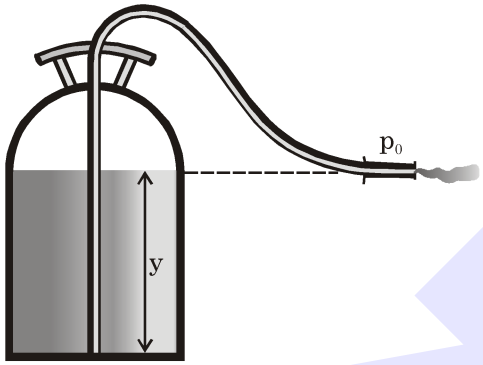
(B) more than half-filled if ρ_C is more than 1.0

(C) half-filled if ρ_C is more than 0.5

(D) less than half-filled if ρ_c is less than 0.5

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5. A certain fire extinguisher consists of a large sealed tank with a thin tube leading from the bottom of the tank to an exit nozzle; the air above the liquid in the tank is at an overpressure p_{tank} with respect to atmospheric pressure p_0. The exit nozzle is at the same level as the top of the liquid; both are a distance y above the bottom of the tank (see figure). The liquid has density ρ. Find an expression for the speed of the stream of liquid emerging from the nozzle.

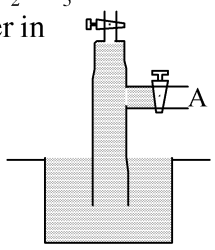


(A) $\sqrt{\frac{2(p_{tank}+p_0)}{\rho}}$ (B) $\sqrt{\frac{2p_{tank}}{\rho}}$ (C) $\sqrt{\frac{(p_{tank}+p_0)}{2\rho}}$ (D) $\sqrt{\frac{p_{tank}}{\rho}}$

6. A tank full of water has a small hole at its bottom. Let t_1 be the time taken to empty first one third of the tank and t_2 be the time taken to empty second one third of the tank and t_3 be the time taken to empty rest of the tank then

(A) $t_1 = t_2 = t_3$ (B) $t_1 > t_2 > t_3$ (C) $t_1 < t_2 < t_3$ (D) $t_1 > t_2 < t_3$

7. Some air is pumped out a tube one end of which is immersed in water. The water in the tube rises above tap A. Now tap A is opened then -

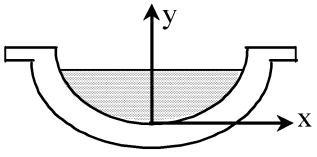


(A) Water will flow from tap and level of water increases
(B) Water will flow from tap A and level of water is same
(C) Water will not flow from tap A
(D) None of these

8. A flat plate moves normally with a speed v_1 towards a horizontal jet of water of uniform area of cross-section. The jet discharges water at the rate of volume V per second at a speed of v_2. The density of water is ρ. Assume that water splashes along the surface of the plate at right angles to the original motion. The magnitude of the force acting on the plate due to jet of water is :–

(A) $\rho V v_1$ (B) $\rho\left(\frac{V}{v_2}\right)(v_1+v_2)^2$ (C) $\frac{\rho V}{v_1+v_2}v_1^2$ (D) $\rho V(v_1+v_2)$

9. A small hole is made at the bottom of a symmetrical jar as shown in figure. A liquid is filled into the jar upto a certain height. The rate of descension of liquid is independent of the level of the liquid in the jar. Then the surface of the jar is a surface of revolution of the curve -

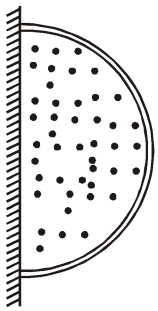


(A) $y = kx^4$ (B) $y = kx^2$ (C) $y = kx^3$ (D) $y = kx^5$

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Multiple Correct Answer Type **4 Q. [4 M (–1)]**

10. A thin walled hemispherical shell of mass m and radius R is pressed against a smooth vertical wall. Through a very small aperture at its top water of density ρ is filled in it completely. Minimum magnitude of force is to be applied to the shell for liquid not to escape from it. Weight of hemispherical shell is mg and total mass of water is $M = \frac{2}{3}\pi R^3\rho$:-



(A) Horizontal component of net force is $\frac{1}{2}\pi R^3\rho g$.

(B) Net external force is $Mg\sqrt{\frac{9}{4}+\left(1+\frac{m}{M}\right)^2}$

(C) Vertical component of force is (M + m)g

(D) Direction of net force is given by $\theta = \text{arc tan}\ \frac{(m+M)}{3M}$, where θ is from horizontal direction.

11. A cubical block is floating in a liquid with one third of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of g/2 :-

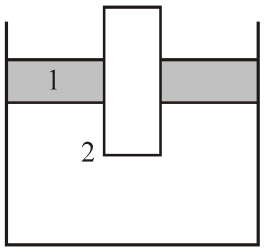
(A) the fraction of volume immersed in the liquid will change.

(B) the buoyancy force on the block will change.

(C) the buoyancy force will increase by 50 percent.

(D) the pressure in the liquid will increased.

12. A cylinder is floating in two liquids as shown in figure.



Choose the correct options

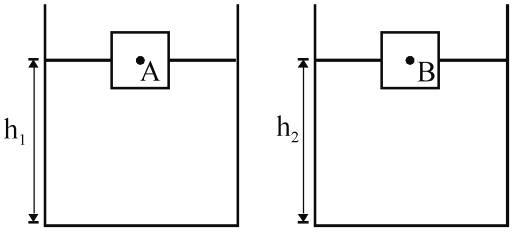
(A) net force on cylinder by liquid 1 is zero.

(B) net force on cylinder by liquid 1 is non-zero.

(C) net force on cylinder by liquid 2 is equal to the upthrust.

(D) net force on cylinder by liquid 2 is more than the upthrust.

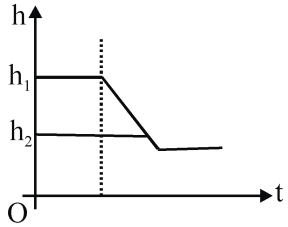
13. Two ice cubes of side 10 cm, having cavity of volume 20 cm^3 at centre of cube but filled with different materials A and B respectively. The specific gravity of material A is 4.9 and specific gravity of material B is 1.9. Now these cubes are placed in two different vessels of same base area as shown in figure. The water level before putting blocks in vessels are same. Assume that ice melts uniformly from all sides and with same constant rate in both the vessels. (specific gravity of ice = 0.9). In (C) and (D) graph showing the variation of heights of water-level in two vessels with time. Choose the correct statement :



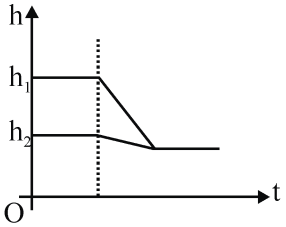
(A) the ratio of initial submerged volumes of the blocks containing A and B is $\frac{49}{46}$

(B) the ratio of initial submerged volumes of the blocks containing A and B is 49/19

(C)



(D)

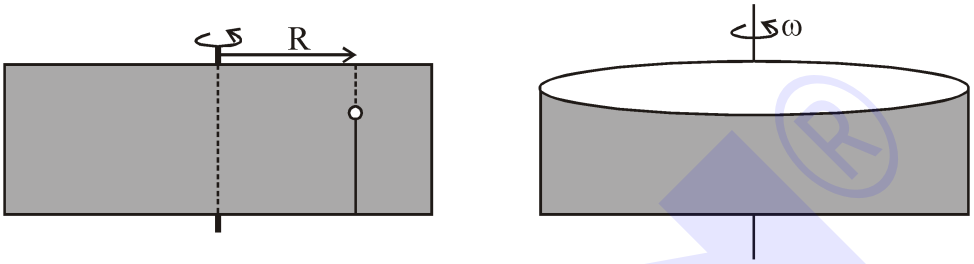


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Linked Comprehension Type **(1 Para × 2Q, 1 Para × 3Q.) [3 M (-1)]**
(Single Correct Answer Type)

Paragraph for Question No. 14 and 15

A closed cylindrical container with a vertical axis is completely filled with liquid (ρ_ω). A plastic bead of density ρ_{body} and volume V is placed at distance R from the axis and is anchored to the bottom of the container by a thin thread of length ℓ. If as a result of the containers revolutions, the beads sink by h and final distance of bead from axis is r. (In the final state the total content of the container rotates at the same angular speed.)



14. What is force exerted by liquid on the bead ?

(A) ρ_ω V g

(B) $(\rho_\omega - \rho_{body})$ V g

(C) $(\rho_\omega - \rho_{body})$ V $\sqrt{g^2 + (\omega^2 r)^2}$

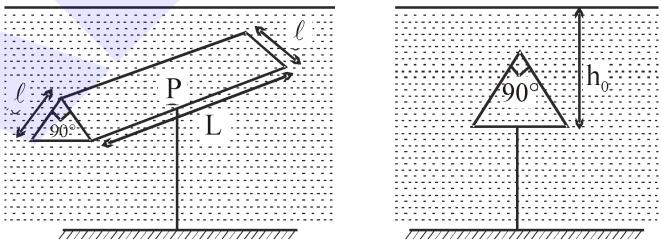
(D) ρ_ω V $\sqrt{g^2 + (\omega^2 r)^2}$

15. What is ω :-

(A) $\omega = \sqrt{\frac{(R+r)g}{(\ell+h)r}}$ (B) $\omega = \sqrt{\frac{(R-r)g}{(\ell-h)r}}$ (C) $\omega = \sqrt{\frac{(R+r)g}{(\ell-h)r}}$ (D) $\omega = \sqrt{\frac{(R-r)g}{(\ell+h)r}}$

Paragraph for Question No. 16 to 18

A prism shaped styrofoam of density $\rho_{styrofoam} < \rho_{water}$ is held completely submerged in water. It lies with its base horizontal. The base of foam is at a depth h_0 below water surface and atmospheric pressure is P_0. Surface is open to atmosphere. Styrofoam prism is held in equilibrium by the string attached symmetrically. (Take : $\rho_{styrofoam} = \rho_f$; $\rho_{water} = \rho_w$).



16. Net force exerted by water on the styrofoam is

(A) $\sqrt{2}\rho_w g\ell^2 L$ (B) $2\rho_w g\ell^2 L$ (C) $\rho_w g\frac{\ell^2 L}{2}$ (D) $\rho_w g\ell^2 L$

17. Magnitude of force on any one of the slant face of styrofoam is

(A) $(P_0 + \rho_w g(h_0 - \sqrt{2}\ell))L\ell$

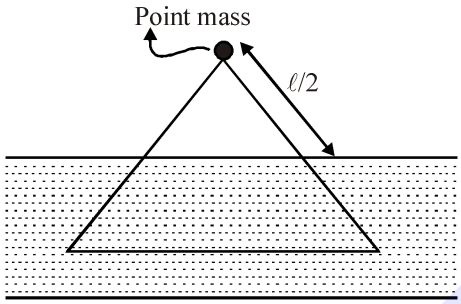
(B) $(P_0 + \rho_w g(h_0 - \frac{\ell}{\sqrt{2}}))L\ell$

(C) $(P_0 - \rho_w g(h_0 - \frac{\ell}{\sqrt{2}}))L\ell$

(D) $(P_0 + \rho_w g(h_0 - \frac{\ell}{2\sqrt{2}}))L\ell$

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18. Now string is cut and styrofoam is allowed to come to surface. A point mass is to be placed symmetrically on the upper surface of styrofoam such that it is in equilibrium with its base in horizontal plane. In equilibrium position styrofoam has half of its slant length submerged. Surface tension of water is T, contact angle is 0°. Determine mg to achieve equilibrium.



(A) $\frac{\rho_f gL\ell^2}{2}+\frac{3}{4}\rho_w gL\ell^2-2T[L+\ell]$

(B) $\frac{-\rho_f gL\ell^2}{2}+\frac{3}{8}\rho_w gL\ell^2-\sqrt{2}T[L+\ell]$

(C) $\frac{-\rho_f gL\ell^2}{2}+\frac{3}{8}\rho_w gL\ell^2-\frac{T[L+\ell]}{\sqrt{2}}$

(D) $\frac{-\rho_f gL\ell^2}{2}+\frac{3}{8}\rho_w gL\ell^2+\sqrt{2}T[L+\ell]$

SECTION-II

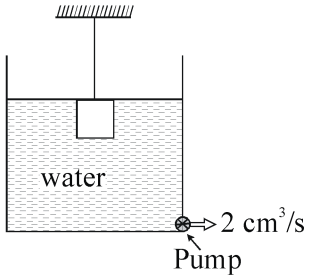
Numerical Answer Type Question **6 Q. [3(0)]**

(upto second decimal place)

19. A U-tube filled with a liquid of volumetric coefficient of 10^{-5}/°C lies in a vertical plane. The height of liquid column in the left vertical limb is 100 cm. The liquid in the left vertical limb is maintained at a temperature = 0°C while the liquid in the right limb is maintained at a temperature = 100°C. Find the difference (in cm) in levels in the two limbs.

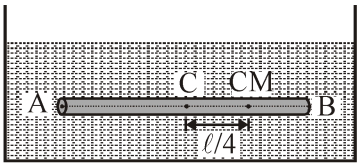
20. An expansible balloon filled with air floats on the surface of a lake with 2/3 of its volume submerged. How deep must it be sunk in the water so that it is just inequilibrium neither sinking further nor rising? It is assumed that the temperature of the water is constant & that the height of the water barometer is 9 meters.

21. Figure shows a cubical block of side 10 cm and relative density 1.5 suspended by a wire of cross sectional area 10^{-6} m^2. The breaking stress of the wire is 7×10^6 N/m^2. The block is placed in a beaker of base area 200 cm^2 and initially i.e. at t = 0, the top surface of water & the block coincide. There is a pump at the bottom corner which ejects 2 cm^3 of water per sec. Find the time (in sec) at which the wire will break.



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22. A non uniform cylinderical rod of mass 5kg, length 2m and radius 0.1m is having its center of mass at a distance $\frac{1}{2}$m from the centre and lying on the axis of the rod. The rod is kept in a liquid of uniform density 1400 kg/m^3. The moment of inertia of the rod about the centre of mass is 4 kg-m^2. Find the angular acceleration (in SI unit) of point A of the rod just after the rod is released from the position shown in figure.



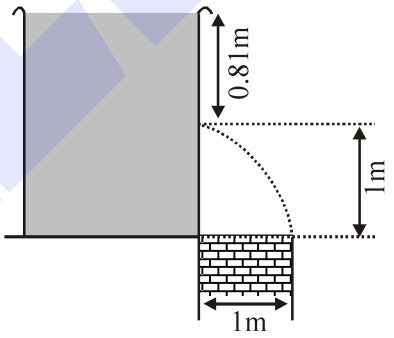
23. A horizontal pipeline carries water in a streamline flow. At a point along the pipe, where the cross-sectional area is 10 cm^2, the water velocity is 1 m/s and the pressure is 4000 Pa. The pressure of water at another point where the cross-sectional area is 5 cm^2, is..... Pa. (Density of water = 10^3 kg. m^{-3}).

24. A block of wood is floating in water such that 1/3rd of it is submerged in water when the same block is floated in alcohol, 1/2 of it's volume is submerged. Now a mixture of water and alcohol is made taking equal volume of both and block is floated in it. What is the % of it's volume that is now submerged.

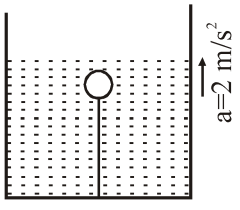
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

25. For the arrangement shown in the figure. The time interval in seconds after which the water jet ceases to cross the wall is found to be $(\alpha)^3$. Area of the cross section of the tank $A = \sqrt{5}$ m^2 and area of the orifice a = 32 cm^2. [Assume that the container remaining fixed]. Find the value of α. (Take g = 10 m/s^2)

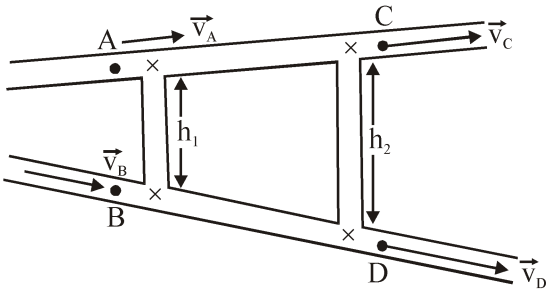


26. A solid sphere of mass m = 2 kg and specific gravity S = 0.5 is held stationary relative to tank filled with water. The tank is accelerating upward width accelration 2 m/s^2. The tension in the string in found to be 8 n. Find the value of n. Given density of water = 1000 kg m^{-3} and g = 10 m/s^2.



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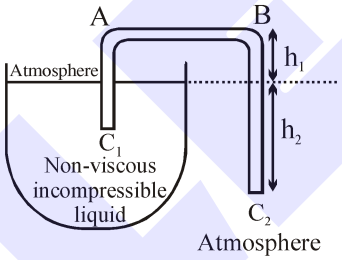
27. An ideal fluid is flowing in two pipes of same cross-sectional pipes area. Both the pipes are connected with two vertical tubes, of length h_1 and h_2 as shown in figure. The flow is stream line in both pipes. If velocity of fluid at A, B, and C are 2 m/s, 4 m/s and 4m/s respectively, the velocity of fluid at D (in m/s) is -



SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0))]**

28. An evacuated thin tube has two tight caps C_1 and C_2 at its ends. The tube is adjusted as shown in the figure below. Atmospheric pressure = P and density of the liquid = d, Neglect surface tension of liquid. Match the entities of columns

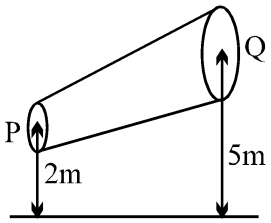


Column-I	**Column-I**
(A) $h_1 > \frac{P}{dg}, h_2 = 0$	(P) If C_2 is opened first and then C_1 is opened, liquid level in the left arm of the tube will be on level with liquid in the pot.
(B) $h_1 > \frac{P}{dg}, h_2 > 0$	(Q) If C_1 is opened with C_2 closed, liquid will rise in the left arm but tube will not be fully filled.
(C) $h_1 < \frac{P}{dg}, h_2 > 0$	(R) If C_1 is opened with C_2 closed, tube will be fully filled with liquid.
(D) $h_1 + h_2 < \frac{P}{dg}, h_2 > 0$	(S) If C_1 is opened first and then C_2 is opened, liquid will flow down continuously.
	(T) After any of the above action pressure at the mid point of tube AB is non-zero

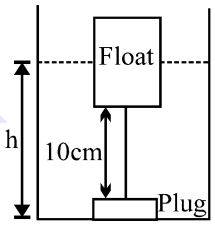
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Subjective Type **2 Q. [4 M (0)]**

29. A non-viscous liquid of constant density 1000 kg/m^3 flows in a streamline motion along a tube of variable cross section. The tube is kept inclined in the vertical plane as shown in the figure. The area of cross section of the tube at two points P and Q at heights of 2 meters and 5 meters are respectively 4×10^{-3}m^2 and 8×10^{-3} m^3. The velocity of the liquid at point P is 1 m/s. Find the work done per unit volume by the pressure and the gravity forces as the fluid flows from point P to Q.



30. A level controller is shown in the figure. It consists of a thin circular plug of diameter 10cm and a cylindrical float of diameter 20cm tied together with a light rigid rod of length 10cm. The plug fits in snugly in a drain hole at the bottom of the tank which opens into atmosphere. As water fills up and the level reaches height h, the plug opens. Find h. Determine the level of water in the tank when the plug closes again. The float has a mass 3kg and the plug may be assumed as massless.



GR # FLUID STATICS & DYNAMICS **ANSWER KEY**

SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (A) **2. Ans. (A)** **3. Ans. (A)** **4. Ans. (A)** **5. Ans. (B)**
6. Ans. (C) **7. Ans. (C)** **8. Ans. (B)** **9. Ans. (A)**

Multiple Correct Answer Type **4 Q. [4 M (–1)]**

10. Ans. (B, C) **11. Ans. (B,C,D)** **12. Ans. (A,D)** **13. Ans. (A,C)**

Linked Comprehension Type **(1 Para × 2Q, 1 Para × 3Q.) [3 M (-1)]**

(Single Correct Answer Type)

14. Ans. (D) **15. Ans. (B)** **16. Ans. (C)** **17. Ans. (D)** **18. Ans. (B)**

SECTION-II

Numerical Answer Type Question **6 Q. [3(0)]**

(upto second decimal place)

19. Ans. 0.1 **20. Ans. 4.5** **21. Ans. 100** **22. Ans. 110** **23. Ans. 2500**
24. Ans. 40

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

25. Ans. 5 **26. Ans. 3** **27. Ans. 2**

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

28. Ans. (A) PQ (B) PQ (C) PRST (D) PRST

Subjective Type **2 Q. [4 M (0)]**

29. Ans. + 29625 J/m^3, – 30000 J/m^3 **30. Ans.** $h_1 = \frac{2(3+\pi)}{15\pi} = 0.26$; $h_2 = \frac{3+\pi}{10\pi} = 0.195$

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